Problems Overview

Introduction

The Port of Dover is the UK's busiest international ferry port, linking southeast England to France across a 33 km route [1]. It handles around 9 million passengers and over 2 million freight vehicles annually [2], with sailings departing approximately every 30 minutes [3]. The port facilitates around £144 billion of trade each year and carries roughly 31% of all seaborne HGV traffic. Its fast turnaround operations generate an estimated £3 billion in annual transport cost savings and support substantial economic activity in Kent [2].

Thoughts. Dover operates a high-throughput, low-buffer system where stochastic vehicle arrivals meet fixed ferry schedules and capacity-limited border processing. Because utilisation frequently approaches system capacity, even small fluctuations in arrivals or processing times can trigger disproportionate delays. Congestion is therefore not merely a result of disruption but a structural feature of a tightly coupled, near-saturated system.

Stakeholders

- **Dover Harbour Board**: Oversees port management and operations [4].
- **Ferry Operators**: P&O Ferries, DFDS, and Irish Ferries provide cross-channel services [3].
- **Border Agencies**: UK Border Force and French Police Aux Frontières conduct juxtaposed checks [5].
- **Local and Transport Authorities**: Kent Police, National Highways, and the Kent Resilience Forum coordinate traffic and resilience planning [6].

Queuing Process

Vehicles pass through check-in and passport control, before entering marshalling lanes for boarding [7]. Traffic flow is supported by real-time monitoring systems using BlipTrack[1] [8]. During peak demand or disruption, additional measures help manage congestion:

- **Dover TAP**: Temporarily holds HGVs on the A20 to prevent congestion in the town centre [9].
- **Operation Brock**: Allocates sections of the M20 as controlled queuing space during severe disruption [6].

Bottlenecks

Delays arise when arrival rates exceed processing capacity, a situation exacerbated by weather-related disruption, ferry delays, or extended post-Brexit passport checks. Reported processing times range from roughly 90 seconds per car to up to 20 minutes per coach [10]. These constraints have produced severe delays, including six-hour queues for cars in 2022 and up to 18-hour queues for coaches in 2023 [11, 12].

[1] BLIP Systems' technology to anonymously detect Bluetooth/Wi-Fi signals for real-time traffic analysis

Parameter	Meaning	Public Availability	Source / Estimation Method	Impact
$\lambda(t)$	Time-dependent arrival rate of vehicles (vehicles per hour)	Partial	Dover annual volumes; DfT hourly traffic counts (proxy); requires estimation	Critical
μ	Average service rate per booth (vehicles per hour per booth)	No	Inferred from observed delays or RoRo processing studies	Very critical
$c(t)$	Number of booths open at time t (decision variable)	Partial	Total installed booths public; operational schedules not public	High
w	Cost of operating one booth per hour (£)	No	Approximated from Border Force pay scales	Moderate
T	Length of planning horizon (hours)	Yes	Defined by analyst (e.g., 24 hours)	Low
$W_q(t)$	Expected waiting time (time per vehicle)	Derived	Computed from $M/M/c$ model	High
$L_q(t)$	Expected queue length (number of vehicles in the queue)	Derived	From Little's Law $L_q = \lambda W_q$	High
$\rho(t)$	Utilisation $\lambda(t)/(c(t)\mu)$	Derived	From model parameters	Very critical

Table 1: Model parameters, definitions, public data availability, and estimated impact on model accuracy.

Parameters labelled "critical" or "very critical" are those for which estimation accuracy strongly affects the model's predictions and optimisation outcomes. Small errors in these values can significantly distort queueing behaviour or recommended staffing levels. Since waiting times in multi-server queues grow nonlinearly as utilisation approaches one, small errors in estimating $\lambda(t)$ or μ can significantly distort predicted delays. By contrast, factors such as ferry capacity influence marshalling but play a secondary role in the immediate queueing dynamics at the border control stage.

Mathematical Formulation

System Definition And Units

To model congestion at Dover's border-control stage, we consider a multi-server service system with the following quantities:

- $\lambda(t)$: vehicle arrival rate at time t (vehicles per hour)
- μ: per-booth service rate (vehicles per hour per booth)
- $c(t)$: number of active booths (dimensionless, integer)

Throughout the model, time is measured in hours and flows in vehicles per hour. Total processing capacity at time t is therefore $c(t)\mu$.

Queue Performance Under Fixed Staffing

Given $(\lambda(t), \mu, c(t))$, system utilisation is

$$\rho(t) = \frac{\lambda(t)}{c(t)\mu},$$

which represents the fraction of available processing capacity being used. When $\rho(t)$ approaches one, waiting times grow rapidly.

Under an $M/M/c$ approximation, the expected waiting time in queue is

$$W_q(t) \approx \frac{P(wait \mid t)}{c(t)\mu - \lambda(t)},$$

and the corresponding expected queue length follows from Little's Law:

$$L_q(t) = \lambda(t) W_q(t).$$

These expressions describe the operational performance for any fixed staffing level $c(t)$.

Stability And Robustness Requirements

To prevent unbounded queue growth, processing capacity must exceed demand:

$$c(t)\mu > \lambda(t).$$

Since $c(t)$ must be integer-valued, a minimal feasible level is

$$c(t) = \left\lceil \frac{\lambda(t)}{\mu} \right\rceil.$$

Because systems operating at high utilisation are sensitive to short-term fluctuations, a robustness requirement such as

$$\rho(t) \leq 0.85$$

is often imposed in practice, yielding

$$c(t) \geq \frac{\lambda(t)}{0.85\,\mu}.$$

Operational Decision Model

In practice, staffing decisions are made over discrete time intervals (e.g. hourly or by ferry-departure windows). Let $t = 1, \dots, T$ index these periods, with arrival rates λ_t and staffing levels c_t.

A discrete-time staffing cost model is therefore

$$\min_{c_t} \sum_{t=1}^{T} w\, c_t,$$

where w is the cost of operating one booth for one time period.
To ensure stability,

$$c_t\mu \geq \lambda_t, \qquad c_t \in Z_{\geq 0}.$$

If a robustness requirement is imposed, the constraint becomes

$$c_t\mu \geq \frac{\lambda_t}{0.85}.$$

This formulation captures the trade-off between staffing levels and congestion risk while aligning directly with the discrete operational decisions made at the port.

Managerial Implications

The queueing model highlights that congestion at Dover is driven by the balance between the arrival rate of vehicles and the processing capacity provided by active booths. The utilisation factor,

$$\rho(t) = \frac{\lambda(t)}{c(t)\mu},$$

quantifies operational stress: as utilisation approaches one, waiting times increase sharply and queues become unstable.

The booth-capacity rule,

$$c(t) = \left\lceil \frac{\lambda(t)}{\mu} \right\rceil,$$

offers stakeholders a practical decision tool. Adjusting the number of open booths in response to arrival-rate fluctuations enables the port to stabilise utilisation and prevent queue growth. Enhancing the effective service rate μ,through digital pre-clearance, simplified documentation checks, or staffing efficiencies, reduces required capacity and improves system robustness.

These insights inform staff scheduling, resource allocation, and day-to-day operational planning. Opening booths ahead of predictable surges, or mitigating inflow through upstream traffic management measures, such as Dover TAP or coordinated time-slot management, can shift operations back into a stable regime. Although the model abstracts from complexities such as weather, ferry cancellations, and inspection variability, it identifies the primary operational levers shaping congestion and provides a structured foundation for more detailed analysis when richer data become available.

References Cited

[1] NASA Earth Observatory, “The strait of dover.” https://www.earthobservatory.nasa.gov/images/6840/the-strait-of-dover, 2006. Accessed: 2025.

[2] D. H. Board, “Key facts and economic impact.” https://www.portofdover.com/about/key-facts/, 2024. Accessed: 2025.

[3] Port of Dover, “Ferry services.” https://www.portofdover.com/ferry/, 2025. Accessed: 2025.

[4] Port of Dover, “The board.” https://www.portofdover.com/corporate/about/the-board/, 2025. Accessed: 2025.

[5] UK Home Office, “The uk’s juxtaposed border controls.” https://homeofficemedia.blog.gov.uk/2023/04/06/the-uks-juxtaposed-border-controls/, 2023. Accessed: 2025.

[6] Kent Resilience Forum, “Traffic management faqs.” https://www.kentprepared.org.uk/faqs, 2024. Accessed: 2025.

[7] A. Evans and P. Wills, “Traffic modelling at the port of dover,” tech. rep., University of Kent, 2017. Accessed: 2025.

[8] “Port of dover in europe installs bliptrack technology to predict traffic congestion,” *Ship Technology*, 2015. Accessed: 2025.

[9] National Highways, “Kent corridors to m25 route strategy: Stage 1 evidence report,” tech. rep., 2023. Accessed: 2025.

[10] BBC News, “Off-road sites could be used for dover port queues.” https://www.bbc.com/news/articles/cdeyprjp0kro, 2024. Accessed: 2025.

[11] S. Meredith, “Uk’s dover port declares ‘critical incident’ as travelers face hours-long wait.” https://www.cnbc.com/2022/07/22/uks-dover-port-declares-critical-incident-as-travelers-face-hours-long-wait.html, 2022. Accessed: 2025.

[12] BBC News, “Dover delays as ferry passengers wait for hours.” https://www.bbc.com/news/uk-65143093, 2023. Accessed: 2025.